Fundamental of Artificial Intelligence



Autumn 2026

By: Dr. Meisam Bakhtiari

A Little About Me

Lecturer: Dr. **Meisam Bakhtiari**

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Google Scholar

ORCiD

Linkedin

Research Interests

- ✓ **Smart materials (Shape Memory Polymers, Alloys and Hydrogels)**
- ✓ **Artificial Intelligence and Machine Learning**
- ✓ **Nonlinear Continuum mechanics**
- ✓ **Laser Shock Peening and Surface Improvement**
- ✓ **Hot Deformation (Recovery, and Static/Dynamic Recrystallization)**
- ✓ **Nonlinear Elasticity and Finite Element Method**
- ✓ **Plasticity (Finite Deformation)**
- ✓ **Computational solid mechanics**

Lectures

Class

Time: 13:00 to 15:00 p.m.

Schedule: Sundays

Location: Hashtroudi Building, Room No. 215

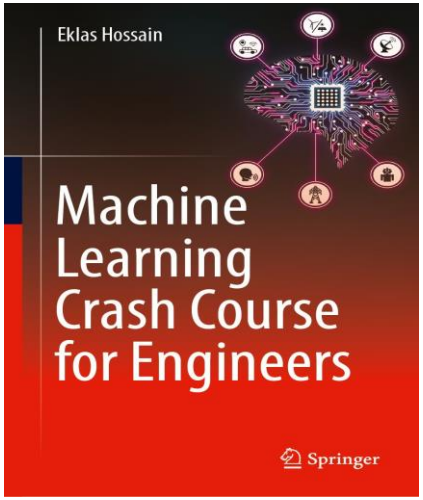
Exam Time & Date

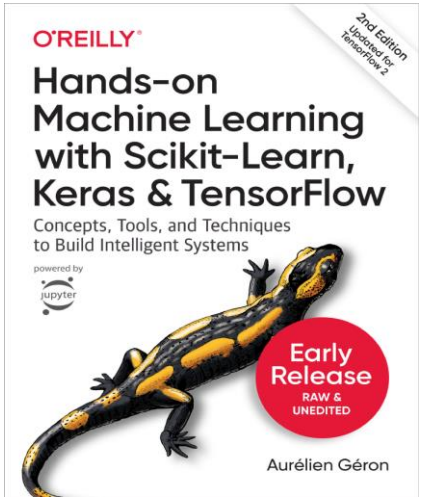
- **Midterm Exam**: **1405/08/24**-13:00 to 15:00 p.m.
- **Final Exam**: **1405/11/17**-09:00 to 11:00 a.m.



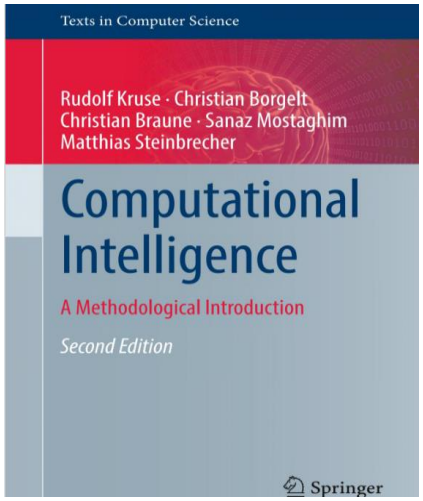
Course References

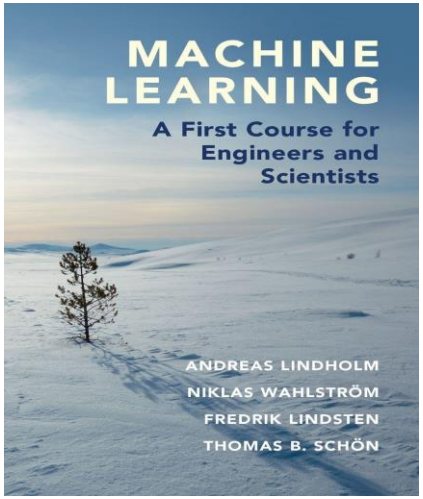
- **Main Ref**: Lecture Slides
- **Required Ref**:





- **Optional Ref:**

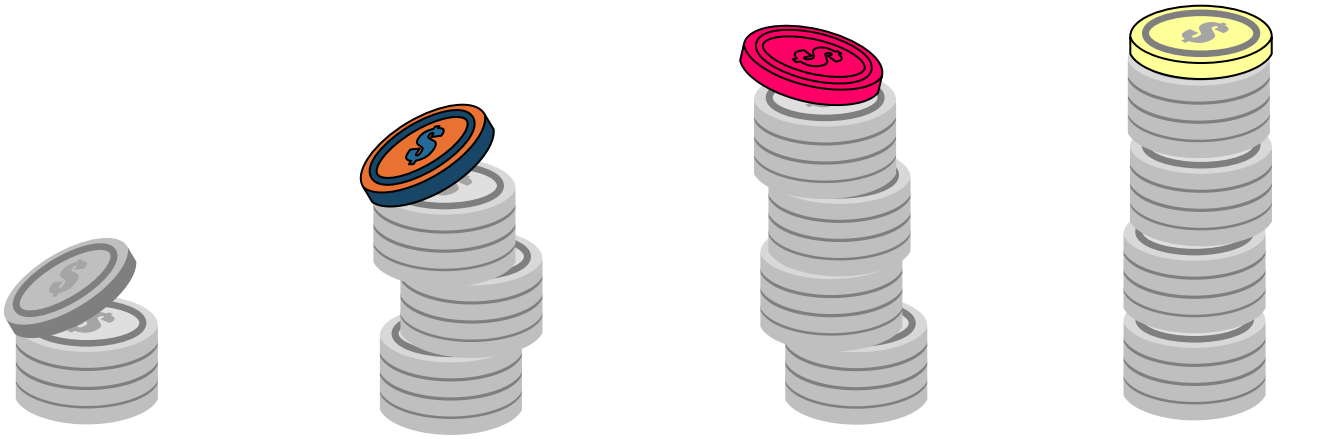




Contents

- Chapter 1: Introduction to Artificial Intelligence and Computational Intelligence
- Chapter 2: Introduction to Python and Libraries
- Chapter 3: Fundamentals of Statistics and Probability
- Chapter 4: Data & Databases
- Chapter 5: Agents
- Chapter 6: Machine Learning Algorithems
- Chapter 7: Supervised Learning Methods
- Chapter 8: Unsupervised Learning Methods
- Chapter 9: Modeling Process
- Chapter 10: Evaluation & Validation
- Chapter 11: Neural Networks
- Chapter 12: Deep Learning
- Chapter 13: Application of AI in Materials Science and Engineering
- Chapter 14: Current Challenges and Future Trends
- Chapter 15: Iran's National Artificial Intelligence Document

Assessment

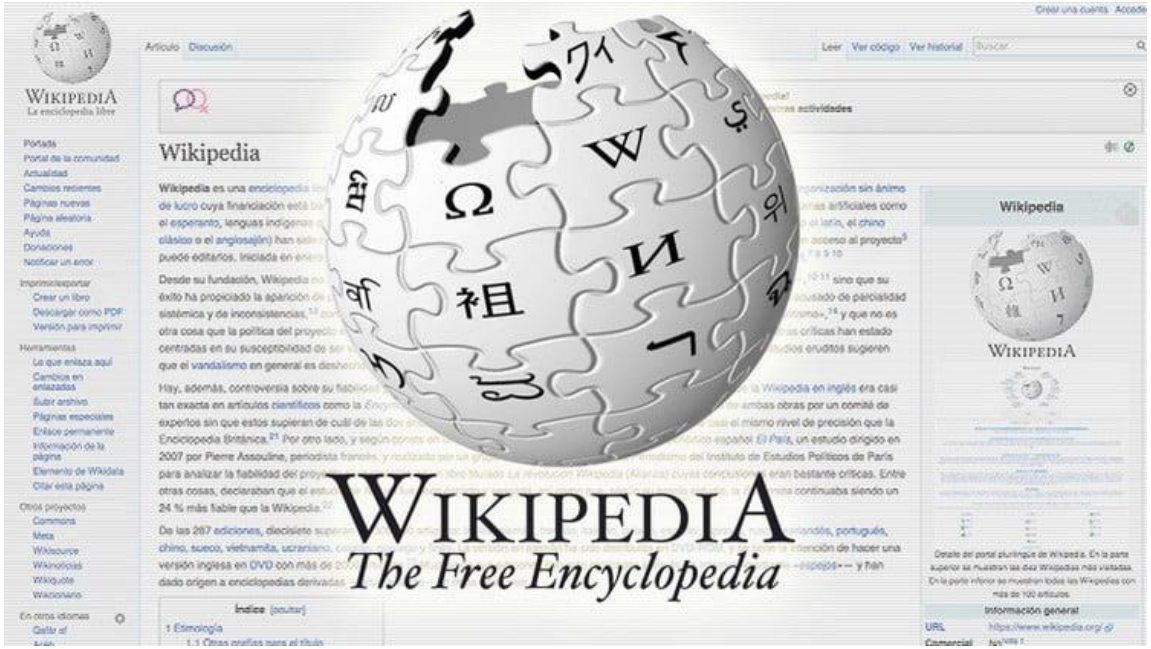


Midterm Exam (20+10%)	Class Activities Mini Projects (15%)	Final Exam (30%)	Main Project (Up to 25%)

- ✓ First Midterm Exam: Chapter 1-5 (17+3)
- ✓ Second Midterm Exam: Chapter 13-15 (17+3)
- ✓ Final Exam: Chapter 6-12 (37+3)
- ✓ Question Type: Multiple-Choice + Descriptive
- ✓ Minor Project: Three or Four Projects
- ✓ Major Project: Three Projects



First Major Project



Project Title: Application of Artificial Intelligent in Materials Science and Engineering or Mechanical Engineering.

Description: 1000-1500 words- 3 Figures- (5-10) References- Link to References and Other Pages.

Languages: English or Persian.